Visiting Alone, but Not Alone: Exploring AI Guides and Peers in Solo VR Museum Visits

ANONYMOUS AUTHOR(S)

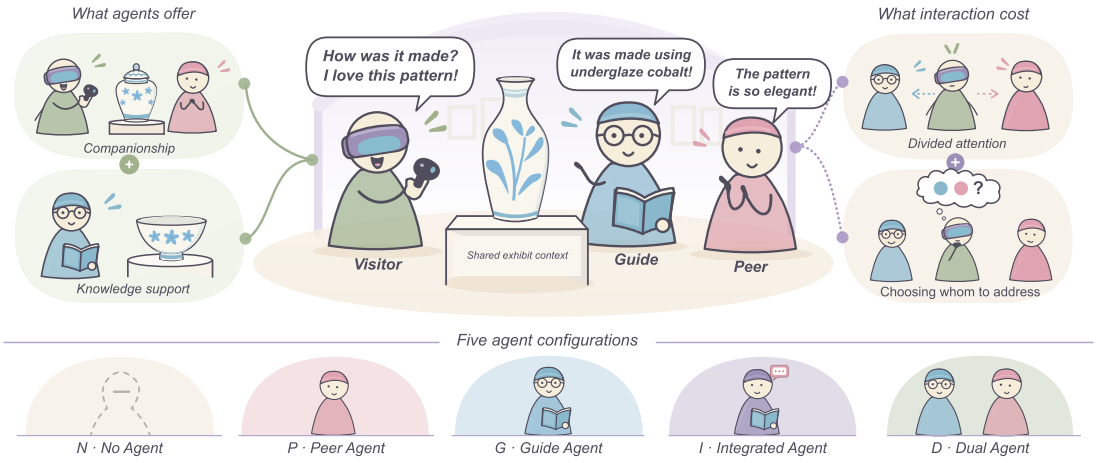


Fig. 1. Conceptual overview of agent roles, interaction costs, and experimental configurations in the VR museum. Agents can provide companionship and knowledge support, but interacting with multiple agents may divide visitors' attention and require them to decide whom to address. We compare five configurations: No Agent (N), Peer Agent (P), Guide Agent (G), Integrated Agent (I), and Dual Agent (D).

In a within-participant study (N = 27), we compared five configurations: No Agent, Peer Agent, Guide Agent, Integrated Agent, and Dual Agents. The Peer Agent received higher companionship ratings than No Agent and the Guide Agent, as well as higher enjoyment ratings than No Agent. The Integrated Agent and Dual Agents also received higher companionship ratings than the Guide Agent, while Dual Agents received higher ratings than the Integrated Agent. However, objective knowledge gain did not differ significantly across configurations. Interviews indicated that peer support facilitated personal expression and artifact interpretation. Integrated Agent responses sometimes failed to address visitors' immediate needs. The distinct identities of the Dual Agents clarified role expectations for some participants but increased attentional and coordination demands. These findings suggest that agent roles in VR museums should be coherent and adaptive, with role separation available on demand.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; • **Computing methodologies** → *Virtual reality*.

Additional Key Words and Phrases: virtual reality museums, embodied conversational agents, large language models, companionship, cultural heritage

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.

Manuscript submitted to ACM

ACM Reference Format:
Anonymous Author(s). 2026. Visiting Alone, but Not Alone: Exploring AI Guides and Peers in Solo VR Museum Visits. 1, 1 (September 2026), 23 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Museums are not only places for learning but also spaces for leisure, relaxation, and social interaction [21, 27]. Visitors often explore exhibitions with family members or friends, discuss artifacts, share personal interpretations, and decide together what to see next [14]. These interactions contribute to enjoyment, reflection, and shared meaning-making. Museum visits are also shaped by visitors' identities and by opportunities to contribute their own voices to interpretation [20]. Companionship is therefore an important part of the museum experience, alongside access to cultural knowledge.

Virtual reality museums extend access to cultural heritage by allowing users to explore immersive exhibitions without geographical constraints [16, 19, 24]. However, VR museum visits are commonly designed as individual experiences. Although multi-user VR can support remote co-visiting, it requires visitors to be available at the same time and may be affected by network latency, technical complexity, and social coordination demands [41]. As a result, the flexibility of solo VR visits may come at the cost of the companionship normally available in physical museums. It remains unclear whether users experience such a companionship gap and whether an artificial companion can meaningfully address it.

LLM-powered agents offer a possible way to provide social interaction without requiring another human visitor. Existing agents in virtual museums, however, are primarily designed as guides that explain artifacts, answer questions, and support navigation [12, 43, 56]. These functions address informational and learning needs but do not necessarily reproduce the informal and reciprocal qualities of visiting with a companion. A Peer Agent could instead explore alongside the user, share reactions, invite discussion, and provide socio-emotional companionship[36]. Yet the value of such a role in solo VR museum visits has received limited attention. We therefore ask: **RQ1: How does Peer companionship affect solo VR museum visits relative to the No Agent and Guide Agent conditions?**

If Peer companionship is valuable, a further question concerns how it should be incorporated into an agent-supported museum experience. One approach is to extend a single Guide Agent with Peer-like functions, such as sharing reactions and encouraging informal discussion. Another is to introduce a separate Peer Agent alongside the Guide [1, 42, 58]. Both approaches combine informational guidance with socio-emotional support, but they organize these functions differently. We therefore ask: **RQ2: How does adding a Peer role to Guide support affect the VR museum experience?**

Separating the Guide and Peer roles may make their functions easier to understand. The Guide can provide reliable information, while the Peer can maintain an informal and socially equal relationship with the user. This separation may strengthen role clarity, perceived complementarity, and social presence [64]. However, interacting with two agents may also require users to divide their attention, decide whom to address, follow additional conversational turns, and manage greater social and coordination demands [17, 33]. The value of the Dual Agents configuration therefore depends on whether the complementary support provided by the two agents justifies these additional costs. This leads to our third question: **RQ3: What are the benefits and costs of combining Guide and Peer roles in one agent versus separating them across two agents?**

To answer these questions, we developed an LLM-powered agent system for solo visits to a VR museum and compared five conditions: No Agent (N), Peer Agent (P), Guide Agent (G), Integrated Agent (I), and Dual Agents (D). The No Agent condition represents an unsupported solo visit. The Peer and Guide conditions emphasized companionship and knowledge support, respectively. In the Integrated Agent condition, one agent performs both the Guide and Peer roles. In the Dual Agents condition, the roles are assigned to separate Guide and Peer agents. This design allows us to examine

the value of Peer companionship, the effect of adding Peer support to a Guide, and the consequences of integrating or separating the two roles.

We evaluate the five conditions in terms of visit convenience, engagement, learning, companionship, and overall experience. We also examine cognitive, attentional, social, and conversational coordination burdens. Together, these measures allow us to determine not only whether Peer support improves solo VR museum visits, but also how its implementation shapes the balance between social benefits and interaction costs.

This work makes three contributions:

- We investigate companionship as a distinct need in solo VR museum visits and distinguish Peer companionship from conventional Guide support.
- We compare five agent configurations to examine how emphasizing, combining, and separating Peer and Guide functions shapes the museum experience.
- We examine the benefits and interaction costs of combining Guide and Peer roles in one agent versus separating them across two agents.

2 Related Work

2.1 Immersive Learning and Social Experience in Virtual Museums

VR museums can enhance presence, engagement, and learning through spatial presentation, reconstructed contexts, and interaction with virtual artifacts [15, 29, 40, 63]. Prior work has also improved visual interaction, navigation, and learning efficiency [30], but has primarily evaluated information acquisition, task performance, and learning outcomes. In physical museums, co-visitors coordinate attention and jointly construct interpretations [66]. Solo VR visitors may therefore lack companionship and emotional exchange [55]. Existing systems enable remote co-visiting [9] and explicitly designed social interaction [59], yet remote systems require synchronous companions and agent-based work rarely treats companionship as a distinct and sustained role. We therefore investigate whether an AI companion can accompany solo visitors, respond to their feelings, and support their social and emotional needs throughout a VR museum visit.

2.2 Roles and Functions of AI Agents in Cultural Heritage

AI agents have been widely used in museums. They have evolved from information-delivery interfaces into interactive roles that perform educational, service, narrative, and creative functions. Previous reviews describe virtual humans in cultural heritage as guides, educators, storytellers, companions, and social mediators [47]. Based on their goals and degree of agent initiative in user interaction, these roles can be classified as Tutors, Guides, On-Demand Assistants, Role-Players, Peers, and Co-Creators. Tutors support users' understanding in VR museums by providing knowledge-based explanations and answering questions [32, 54]. Guides help users visit museums with clear purposes through navigation and recommendations [74]. On-Demand Assistants provide answers and support when users make requests [13, 23]. Role-Players act as specific people or objects and use a first-person perspective to construct cultural contexts within particular situations. Peers interact with users from an equal position by sharing discoveries, engaging in casual discussions, and responding to users' emotions. Co-Creators work with users to produce artworks or reconstruct personal narratives. Among these roles, Peers that explicitly provide socio-emotional support have rarely been designed for museum experiences.

These roles are not mutually exclusive [1, 58]. The same agent can act as a Tutor by providing knowledge, as a Guide by directing the visiting route, and as a Role-Player by reconstructing historical or cultural contexts. Multifunctional agents can provide users with different forms of support, but this does not guarantee that their different identities are compatible. Tutors have epistemic authority, while Guides need to balance direction with users' freedom to explore. On-Demand Assistants should avoid interrupting a visit when users have not requested help, whereas Peers need to communicate with users from an equal position. Combining these roles within a single agent reduces the number of interactive entities that users need to manage. However, unclear identities and degree of agent initiative may negatively affect the visiting experience. Existing research has mainly focused on single agents that perform educational functions. Few studies have examined either a single agent that combines educational and companionship functions or multiple agents that perform these functions separately. Therefore, whether a dual agent system with clearly defined roles and responsibilities can improve role clarity and the overall visiting experience remains to be examined [73].

2.3 Peer Agents for Socio-Emotional Support

Social Agency Theory and the Computers Are Social Actors (CASA) paradigm suggest that users respond socially to humanlike cues such as language, gaze, and bodily behavior [11, 44, 46, 52]. Relational agent research shows how repeated interaction and trust-building cues can support longer-term human agent relationships [3, 5, 53]. Building on these ideas, a Peer Agent adopts an equal-status position and provides socio-emotional responses. Compared with tutors and guides, which mainly instruct or direct users, a Peer Agent reduces hierarchical distance and strengthens the sense of sharing the visit with another social actor [36, 48].

Existing museum AI agents have used social responses and companion-like communication to sustain user interaction [4, 6]. However, knowledge delivery remains their primary function. Communication based on visitors' interests and emotions is usually an added feature rather than a distinct role. Evidence from conventional museums cannot directly answer the same question in VR museums. In a virtual environment, an agent's speech and movements may compete with exhibits for the user's attention [25, 60]. Our study therefore examines whether visitors need an explicitly defined Peer role in VR museums and when this need arises. Even if such a need is established, it remains unclear whether the Peer and Guide roles should be performed by one agent or assigned to separate agents.

2.4 Role Specialization in Multi-Agent Systems: Complementarity and Burdens

Multi-agent conversational systems distribute responsibilities across agents and coordinate specialized roles [2]. Prior research has also examined speaker coordination and role-aware listening in embodied multi-party dialogue [64, 69]. In a VR history-education setting, Zhu et al. [76] investigated an LLM-powered pedagogical agent that switched between knowledge-oriented roles and actions. Their findings suggest that role switching can provide complementary support but may also distract users and reduce experiential consistency.

Closer to our setting, Su et al. [61] developed *SimViews*, which uses visitor agents with different professional identities and participation patterns to present diverse perspectives on museum artifacts. Whereas SimViews focuses on viewpoint diversity and conversational participation, our study examines companionship as a primary Peer responsibility and compares combining Guide and Peer roles in one agent with separating them across two agents.

At the same time, multiple agents may increase attentional and conversational demands, although their effects depend on task design and orchestration [57, 62]. We therefore compare an Integrated Agent with a Dual Agents configuration to examine the trade-off between complementary support and interaction burden.

3 Methodology

3.1 Prototype Design

3.1.1 Guide and Peer Role Design. In designing the roles of our AI agents, we considered how to provide informational and companionship-oriented support during virtual museum visits. Accordingly, we defined relational roles with distinct functional priorities rather than mutually exclusive behavioral categories: Guide and Peer. Both agents could discuss artifact content and respond socially when appropriate. **(D1.1) Design the Guide as a knowledge-oriented role that communicates information about artifacts and their cultural contexts.** Its primary behaviors include introducing and answering questions about an artifact's visual features, history, and cultural background, as well as suggesting routes through the museum. The Guide prioritizes accuracy, clarity, and relevance and grounds its responses in curated and verified cultural heritage information [31, 39]. Its primary goal is to help users understand the artifacts rather than to sustain companion-like social interaction. **(D1.2) Design the Peer as a co-visitor who provides companionship-oriented support.** Unlike the Guide, which serves as an authoritative source of knowledge, the Peer is positioned as a fellow visitor on relatively equal status with the user. Museum visits with companions commonly involve jointly attending to exhibits, expressing personal reactions, exchanging perspectives, and responding to each other's emotions [14, 38, 45]. Drawing on these real-world companion behaviors, we designed the Peer to share subjective reactions to exhibits, respond to users' emotions and interests, and sustain conversation by commenting on exhibits, asking follow-up questions, and establishing joint attention. The Peer could respond to knowledge-related questions by commenting on visible features and available exhibit information, but did not introduce knowledge beyond those sources. Its primary purpose is to help users feel that they are viewing, responding to, and discussing the exhibits with a companion.

3.1.2 Integrated Agent and Dual Agents Configurations. After defining the primary responsibilities of the Guide and Peer roles, we considered how informational and companionship-oriented support should be embodied at the agent level. Specifically, the two roles could either be combined within a single agent or assigned to two separate agents. To create a more lifelike museum experience, we embodied the agents as virtual human characters. Because the two configurations differ in both role allocation and the number of embodied agents, we treated them as distinct agent configurations and designed them as follows. **(D2.1) Integrate the Guide and Peer roles within a single agent.** In the Integrated Agent configuration, one agent performs the primary responsibilities of both roles and provides different forms of support based on the user's input and the current interaction context. When a user asks a factual question about an artifact, the Integrated Agent provides a knowledge-based answer and adds an affective or socially responsive comment when the conversational context calls for one. When a user expresses a personal interest, opinion, or feeling, the Integrated Agent prioritizes Peer-oriented behaviors by acknowledging the user's response and asking follow-up questions when appropriate. Presenting both functions through a single agent aims to reduce the need for users to shift their attention between multiple interlocutors while maintaining continuity in the conversational context. However, combining epistemic authority and peer identity within one agent may also blur the boundary between the two roles. To mitigate this issue, we constrained the goals and content of each response type: Guide role responses were grounded in curated and verified cultural heritage knowledge, whereas Peer role responses focused on observable features, personal interpretations, and social exchange rather than introducing additional authoritative claims. **(D2.2) Distribute the Guide and Peer functions across two coordinated, role-specialized agents.** In the Dual Agents configuration, the Guide and Peer roles are embodied by two agents with distinct identities and communicative goals. The Guide provides artifact explanations, contextual knowledge, and navigation suggestions, whereas the Peer shares

personal reactions, responds to the user, and sustains co-visiting interactions. Separating the two roles gives each form of support a clearly identifiable social source and may help users develop stable expectations about each agent's role and function. Moreover, presenting the Peer as an agent distinct from the Guide allows us to examine whether the presence of an explicit peer role affects users' sense of companionship. However, interacting with two agents may require users to distribute their attention across multiple interlocutors. We therefore specified clear speaking conditions, role priorities, and turn-taking coordination mechanisms for the two agents. At a conceptual level, the Integrated Agent and Dual Agents configurations provide the same intended Guide and Peer functions. The two conditions implemented these functions through different configurations of embodiment, identity, routing, turn-taking, and agent participation. Accordingly, the I–D comparison evaluates two complete interaction configurations rather than isolating the causal effect of role separation.

3.1.3 Artifact Selection. To meet the needs of the experiment, we designed the virtual museum exhibition to balance thematic coherence with object diversity. We selected 15 East Asian and European ceramic objects dating from approximately 1675 to 1735 from the Porcelain Collection of the Staatliche Kunstsammlungen Dresden (SKD). All 15 objects came from the same collection and digital resource. The selection comprised one unglazed stoneware object from Yixing, China, and 14 porcelain objects produced in China, Japan, or Meissen, Germany. The objects were connected by their shared collection context, related ceramic materials, and defined historical period, while differences in their places of origin, functions, forms, and decoration provided visual and thematic variation.

Building on these commonalities and differences, we organized the 15 objects into five themes. *Forms for Use* compared teapots from three countries, *Images in Transit* examined how Meissen chinoiserie reinterpreted East Asian imagery, *Nature in Porcelain* presented animal figures, *The World in Porcelain* covered religious, travel, and literary narratives, and *Making Wonders* highlighted openwork and relief techniques. Together, the themes connected everyday life, cross-cultural imagination, nature, narrative, and manufacturing while providing visual and thematic variation.

3.2 Implementation

3.2.1 Application Overview. We developed a Meta Quest 3 VR museum application supporting artifact viewing, bilingual exhibit-label reading, and speech interaction with five embodied-agent configurations. In each visit, the system loaded a thematically related artifact set and the assigned agent configuration. Users could explore at their own pace, access exhibit information, and speak with agents whose responses used their input and current viewing context. The following subsections describe the environment, agent embodiment, and dialogue architecture. Fig. 2 summarizes the prototype and experimental materials.

3.2.2 Virtual Museum Environment. The virtual museum consisted of an entrance area and a main exhibition hall. The hall contained three identical display plinths, each presenting one of three thematically related artifacts in each visit. Participants navigated the hall using VR controllers and viewed the artifacts in any order they preferred. The system did not impose a fixed viewing route [41, 67]. In configurations that included the Guide role, the embodied agent could provide optional suggestions about the viewing order, which participants were free to follow or ignore. Each artifact was accompanied by an interactive bilingual label in Chinese and English. The label initially displayed basic information, including the artifact's name, date, place of production, and material, and could be expanded to show a more detailed description. We made the adaptations necessary to present the official 3D models in VR, including adjustments to their coordinate orientation, pivot points, levels of detail, and texture settings. To make smaller artifacts easier to inspect, all models were proportionally rescaled using the same rule so that their longest dimension was approximately 1.25 m

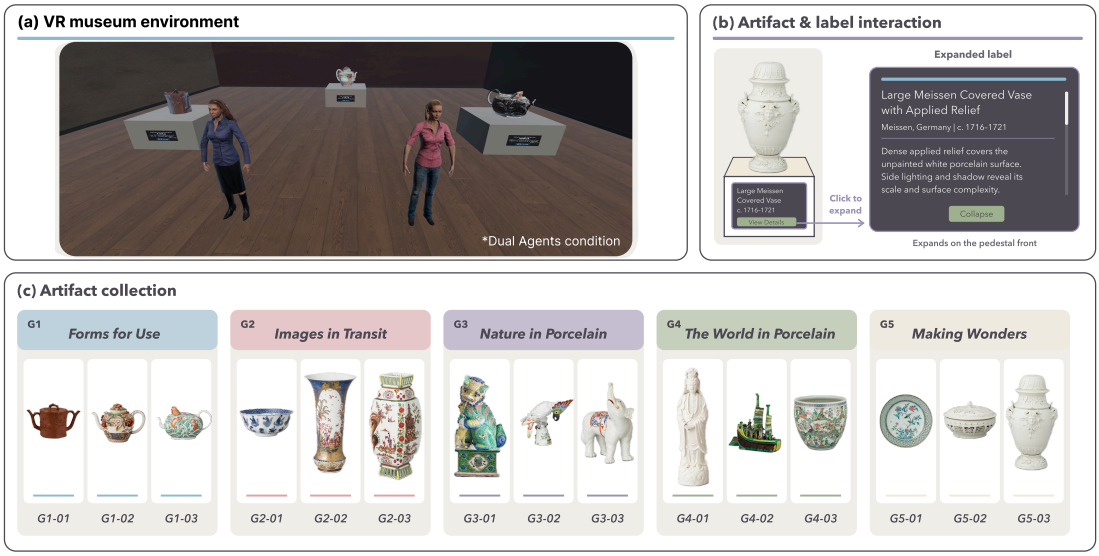


Fig. 2. Overview of the VR museum prototype and experimental materials. (a) A representative scene from the Dual Agents (D) condition. Three artifacts and their exhibit labels are visible in the environment. (b) The exhibit-label interaction: selecting "View Details" expands the bilingual label on the front of the display plinth. (c) The 15 artifacts used in the study, organized into five thematic sets (G1–G5) containing three artifacts each. The assignment of thematic sets to experimental conditions was counterbalanced across participants.

[68]. This adjustment changed only the overall display size while preserving the internal proportions of each model. Before the study, participants were explicitly informed that the virtual display sizes did not represent the artifacts' actual physical dimensions. Apart from the displayed artifacts, the gallery layout, display stands, lighting, label format, and interaction methods remained consistent across all visits.

3.2.3 Agent Embodiment and Interaction. The embodied agents representing the Guide, Peer, and Integrated roles were presented as full-body 3D virtual characters. Their appearances were randomly selected from the Microsoft Rocketbox library, and all characters used the same skeletal rig and animation system [65]. The characters could display standing, listening, thinking, and speaking states. During speech, viseme data drove their lip movements, accompanied by subtle head movements and gestures. At the beginning of each round, the characters introduced themselves using neutral scripts of comparable length. Their voices were spatialized from their positions in the virtual environment, and text captions appeared near the current speaker [10, 11, 13]. We also predefined role-specific content scopes for each artifact. The Guide content covered factual information such as the artifact's date, materials, production techniques, and historical background, whereas the Peer content focused on visible features, personal reactions, and shared viewing.

The Integrated Agent interacted with the user through a single, consistent appearance, voice, and identity. Its responses could combine knowledge-oriented and companionship-oriented content, but the system did not distinguish these functions through role labels, role-switching animations, or other interface cues. In the Dual Agents configuration, the Guide and Peer roles were represented by two separate virtual characters. Each character had an independent spatial audio source and caption position. When one character spoke, its lip movements, head movements, and gestures

were activated, while the other remained in a listening state. The two characters spoke sequentially to prevent their speech, captions, and animations from overlapping.

3.2.4 Dialogue Architecture. Our dialogue system integrates local speech recognition, role-aware speaker assignment, context-grounded response generation, and multimodal embodied-agent output within a real-time VR museum environment (see Fig. 3). Users pressed a VR controller button to start speaking and pressed it again to stop. During this period, the built-in microphone of the Meta Quest 3 captured their speech. The recording was transmitted to a local workstation through a USB connection and transcribed using whisper.cpp 1.9.1 with the multilingual small model (ggml-small.bin) [51]. The resulting text was then passed to the agent-routing module in Unity.

The routing module selected the role prompt used for response generation based on the current agent setup. When one agent was present, the module directly used the corresponding Guide, Peer, or Integrated prompt. When two agents were present, the system selected the primary speaker based on whether the user explicitly addressed a role, the semantic keywords in the response, and the agents' previous participation. Knowledge-oriented questions were primarily assigned to the Guide, whereas responses concerning observations, feelings, and personal preferences were primarily assigned to the Peer. When necessary, the other agent provided a brief, role-consistent follow-up after the primary response [26, 69, 70]. The selected role prompt was then combined with structured information about the current artifact and recent dialogue history. When an exact requested fact was unavailable, the system instructed the agent to redirect the response to the closest supported exhibit information or observable features and to avoid generating unsupported factual claims [31]. The combined input was sent to the DeepSeek Chat Completions API, using the deepseek-v4-pro model to generate a textual response. The generated text was then converted into speech using Azure Cognitive Services Speech SDK 1.51.1. The voices of all three virtual characters were selected from the Azure Speech voice library. Azure Speech returned the audio and corresponding viseme information. The VR application played the spatialized speech, displayed subtitles, and used the viseme data to drive the speaking agent's lip movements.

3.3 User Study

3.3.1 Experimental Conditions. We used a single-factor within-participant design with five conditions: No Agent (N), Peer Agent (P), Guide Agent (G), Integrated Agent (I), and Dual Agents (D). Each participant experienced all five conditions. The condition order and the assignment of the five exhibit-theme packages to the conditions were counterbalanced across participants using Williams sequences to reduce potential order, learning, fatigue, and theme effects [72]. The No Agent condition provided no conversational agent, while retaining the same task structure, exhibit-label format, and access to basic artifact information. Because no conversational interaction was available in this condition, participants could complete the visit after examining the artifacts and exhibit labels. The Peer Agent provided companionship-oriented support by sharing personal impressions, responding to users' feelings, and inviting discussion, but did not provide systematic knowledge explanations. The Guide Agent provided knowledge-oriented support by explaining artifact backgrounds, answering factual questions, and guiding users' observations. The Integrated Agent used one embodied agent to provide both Guide and Peer functions, whereas the Dual Agents condition assigned these functions to two separate embodied agents.In the Dual Agents condition, the system prioritized the agent explicitly addressed by the user. When no agent was specified, knowledge-oriented questions were primarily routed to the Guide, whereas responses concerning observations, feelings, or personal preferences were primarily routed to the Peer. Thus, the Integrated Agent and Dual Agents conditions pursued the same broad goals of providing Guide and Peer support but implemented them through different interaction configurations. The I–D comparison therefore evaluates outcomes

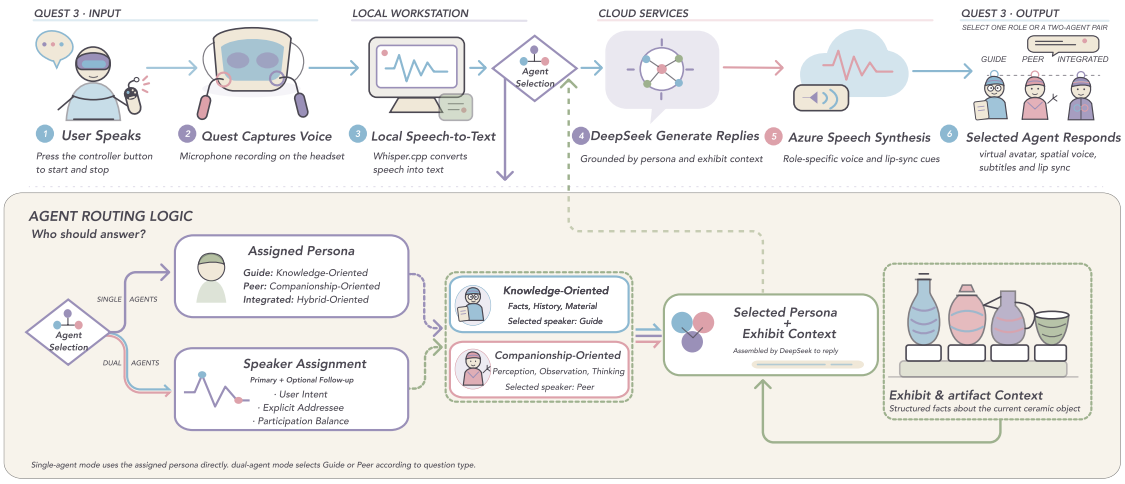


Fig. 3. System pipeline for role-routed, context-grounded voice interaction in the Quest 3 experience. Top: speech input is transcribed locally, grounded in persona and exhibit context, and returned through a selected virtual agent. Bottom: the single-agent configurations use an assigned role prompt, whereas the Dual Agents configuration routes knowledge-oriented questions to the Guide and other inputs, such as observations, feelings, and personal preferences, to the Peer.

associated with these configurations rather than isolating role separation as a single causal factor. Participants were informed that each round would take approximately five minutes, but no minimum duration was enforced. They could end a round once they felt they had completed their visit, and any ongoing agent response was allowed to finish. The exhibit-theme packages were counterbalanced across conditions as described above. Conditions involving agents used the same underlying dialogue-generation and voice-interaction pipeline, while the content and behavior of each agent were constrained by its assigned role.

3.3.2 Participants. We recruited 27 participants through social media (M_{age} = 23.44 years, SD = 2.76, range = 18–30). Participants were required to be at least 18 years old, have sufficient vision and hearing to complete the interactions, and have no health conditions that prevented them from using a virtual reality head-mounted display. We also recorded their academic disciplines, frequency of prior VR use, museum-visiting experience, and familiarity with ceramics. These variables were used to characterize the participants' backgrounds and assess their comparability across conditions. The study was approved by the university's ethics review board, and informed consent was obtained from all participants.

3.3.3 Procedure. Upon arrival at the laboratory, participants were introduced to the study procedure. They then provided informed consent, completed a VR safety screening, and reported their prior VR experience, museum-visiting experience, and familiarity with ceramic artifacts. The researcher then introduced the basic operation of the Meta Quest 3 and provided a short practice session to familiarize participants with navigation, exhibit-label viewing, and voice interaction in the VR environment. Participants next completed a 30-item multiple-choice knowledge pre-test, with all questions based on information presented in the exhibit labels for the 15 artifacts.The formal experiment consisted of five VR museum rounds, each lasting approximately 5 minutes. Participants were required to ask at least three questions in each round, except in the No Agent condition. They completed the five conditions in their assigned order and were free to view the artifacts and interact with the system at their own pace and in their preferred order. After

each round, participants completed six knowledge questions related to the artifacts presented in that round and an experience questionnaire. They then rested for 30–60 seconds before proceeding to the next round. This procedure was repeated until all five rounds were completed. Finally, we conducted a one-on-one semi-structured interview to further understand participants' perceptions of the different agent configurations, companionship experience, and interaction burdens. The entire study lasted approximately 35–50 minutes.

3.3.4 Measures.

Self-Report Measures. After each round, participants rated their perceived knowledge gain (3 items), companionship (3 items), enjoyment (2 items), relaxation (2 items), and overall experience (1 item) on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree). The perceived knowledge and enjoyment items were adapted from the Knowledge/Learning and Engagement dimensions of the Museum Experience Scale, respectively [49]. The companionship items were adapted from the co-presence, mutual awareness, and mutual attention dimensions of the Networked Minds Social Presence Inventory [7], and the relaxation items were adapted from the Restoration Outcome Scale [7, 37]. Overall experience was assessed using one study-specific item. Participants completed single-item manipulation checks applicable to the support functions implemented in each condition. Knowledge Support (MK) was administered in G, I, and D, whereas Companionship Support (MP) was administered in P, I, and D. Because these items were not fully crossed across all agent-present conditions, we treat them as convergent evidence that the intended support was perceived, rather than as a test that the two role functions were mutually exclusive. The Integrated Agent and Dual Agents conditions additionally assessed role clarity and interaction burden, including decision burden, attention switching, and role coordination. After completing all five rounds, participants reported their overall preferences among the five systems and directly compared the Integrated Agent and Dual Agents in terms of role clarity, ease of use, interaction burden, and the benefit–burden trade-off.

Knowledge Measures. We used 30 multiple-choice questions to assess participants' objective knowledge of the 15 artifacts. Each artifact was assessed with one factual question and one question concerning its meaning or context. Before the experiment, participants completed all 30 questions as a pre-test. After each condition, they completed the six questions corresponding to the three artifacts presented in that condition. Correct answers received one point, whereas all other responses received zero points. The overall pre-test score ranged from 0 to 30, and each post-condition knowledge score ranged from 0 to 6.

Behavioral Measures. The VR application automatically recorded the number of exhibit-label openings, gaze duration, the total number of user questions, and the number of questions associated with each artifact in every round. Each record was linked to the participant, round, condition, thematic set, and artifact, and included a timestamp and information about the current artifact. These data supported condition-level and artifact-level behavioral analyses.

3.3.5 Data Analysis. For the quantitative analysis, we examined questionnaire responses, knowledge test scores, and VR logs from 27 participants. Comparisons of questionnaire and knowledge scores across conditions preserved within-participant pairing. After checking data completeness, we averaged the items in each multi-item scale and calculated Cronbach's α separately for each condition to assess internal consistency [18]. Single-item measures retained their original scores. Objective knowledge gain for each round was calculated as the score on the six questions in that round minus the pretest score on the same six questions.

Friedman tests [22] assessed overall differences across configurations for the six measures collected in all five conditions, with Kendall's W [34] reported as the effect size. Holm correction [28] was applied across the six tests. To address the research questions, we compared P–N, P–G, I–G, D–G, and D–I for each common measure. Role clarity and interaction burden were compared only between D and I. These paired comparisons used two-sided Wilcoxon signed-rank tests [71], with rank-biserial correlations [35] reported as effect sizes. Holm correction was applied to the five comparisons within each common measure and separately to the two comparisons for role clarity and interaction burden. The significance level was set to 0.05, and p_{adj} denotes Holm-adjusted p-values within the corresponding family of tests. Exact sign tests were used for sensitivity analyses of the paired comparisons for the common measures. Full computational details and results are provided in the supplementary material. For the five items directly comparing I and D, we conducted two-sided exact binomial tests using only responses that explicitly selected either configuration. These tests assessed whether I and D were equally likely to be selected among respondents who made a choice. Holm correction was applied across the five tests. VR logs were summarized using descriptive statistics, including round duration, the number of voice submissions, and role participation, to describe the interaction process.

To assess implementation fidelity, we coded the interaction logs using a documented codebook. AI assistance generated the initial labels, after which one researcher reviewed every applicable record. The audit covered user intent and routing appropriateness for 149 requests in the Dual Agents condition, and reply function and role adherence for 456 delivered replies in the 503 request analysis set. Routing appropriateness was coded as Yes, Partial, No, or Unclear based on the manually judged user intent and selected speaker. No route in the final coding was classified as Partial. Role adherence was coded as Yes when a reply contained role-consistent behavior, even if it fulfilled only part of the expected role, and as No when it did not. No delivered reply was classified as Partial. The codebook, decision rules, audit summary, and row-level final codes are provided in the supplementary material.

We conducted a thematic analysis of the interview transcripts, iteratively comparing codes and grouping them into themes [8]. Interview recordings were transcribed, and the transcripts were coded, with the research questions guiding the organization of the material. Related codes were compared and grouped into four themes that captured shared and contrasting experiences across participants.

4 Results

This section first reports quantitative results from the questionnaires, knowledge questions, final choice questions, and VR logs, followed by four themes identified through interview analysis. The quantitative results compare performance across configurations, while the qualitative findings help explain participants' experiences and differences across configurations.

4.1 Quantitative Results

All 27 participants completed the questionnaires and knowledge questions in all five configurations. For the multi-item scales used across all configurations, Cronbach's α ranged from 0.733 to 0.934. Scores are reported as means (standard deviations). W denotes Kendall's W, T denotes the smaller rank sum in the Wilcoxon signed-rank test, and r_{rb} denotes the rank-biserial correlation coefficient. All p_{adj} values were adjusted using Holm's method within the corresponding family of tests.

4.1.1 Manipulation and Implementation Checks. Participants generally perceived the support functions intended in each condition. Knowledge Support ratings were 5.81 (1.08), 5.78 (0.85), and 6.04 (0.81) in G, I, and D, respectively.

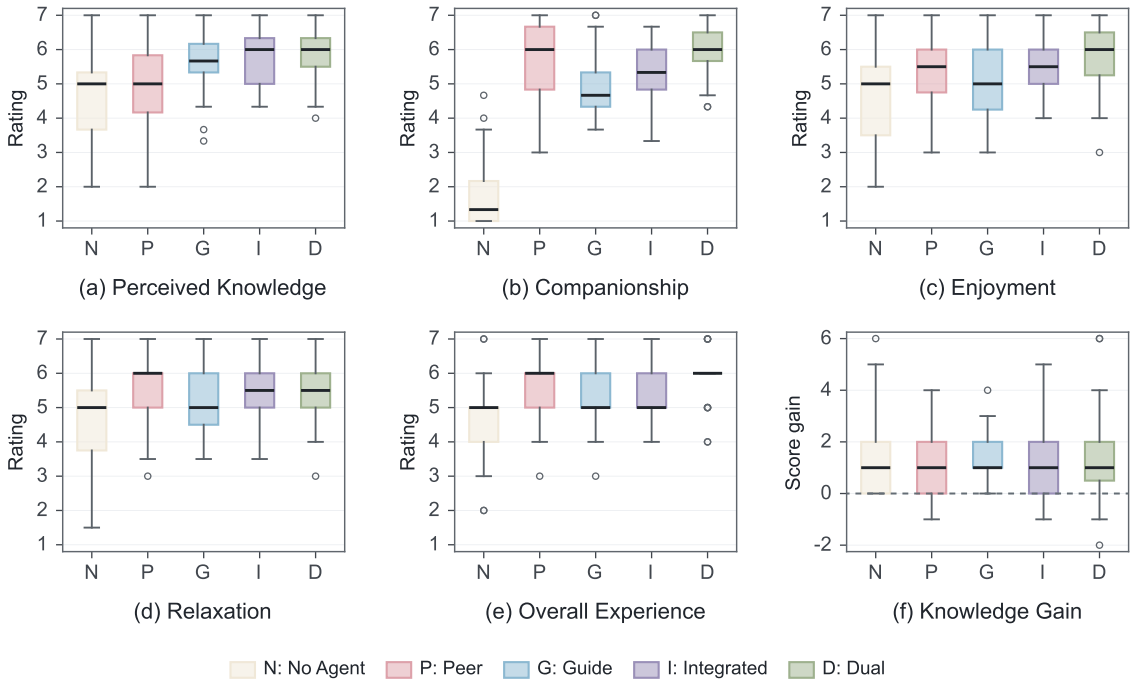


Fig. 4. Distributions of subjective ratings and knowledge gain across five agent configurations. Boxes show the interquartile range, and horizontal lines indicate medians. Whiskers extend to the most extreme observations within 1.5 interquartile ranges of the box edges.Hollow circles indicate outlying observations.The dashed line in (f) indicates zero knowledge gain.

Companionship Support ratings were 5.85 (1.10), 5.26 (1.20), and 5.96 (0.85) in P, I, and D, respectively. These results indicate that the intended forms of support were perceived, but they do not establish discriminant role specificity because MK was not administered in P and MP was not administered in G.

Verified delivery and role realization. Across the four agent-present conditions, 456 of 503 requests produced a delivered reply (90.7%). Thirty-nine requests received no backend response (7.8%), and eight were blocked during delivery (1.6%). In the Dual Agents condition, 136 of 149 requests produced a delivered reply. Human-verified coding classified 119 routes (79.9%) as appropriate, none as partially appropriate, 21 (14.1%) as inappropriate, and 9 (6.0%) as unclear. Among the 136 delivered replies, 95 were delivered by the Guide and 41 by the Peer. Twenty-three participants received at least one direct Peer reply, although 12 received only one. Across all 456 delivered replies in the analysis set, 442 (96.9%) followed the assigned role and 14 (3.1%) did not.

4.1.2 Experience and Knowledge. **Perceived Knowledge.** Scores differed significantly across the five configurations ($\chi^2(4) = 34.67$, $p_{\text{adj}} < 0.001$, $W = 0.321$). Scores for N, P, G, I, and D were 4.80 (1.13), 4.93 (1.22), 5.68 (0.94), 5.78 (0.76), and 5.86 (0.80). Pairwise comparisons showed that P scored lower than G ($T = 56.5$, $p_{\text{adj}} = 0.028$, $r_{\text{rb}} = -0.623$). The remaining four comparisons were not significant (all $p_{\text{adj}} \geq 0.587$).

Companionship. Scores differed significantly across the five configurations ($\chi^2(4) = 70.93$, $p_{\text{adj}} < 0.001$, $W = 0.657$), with the largest effect size among the six common measures. Scores for N, P, G, I, and D were 1.85 (1.04), 5.73 (1.16),

4.94 (0.90), 5.31 (0.91), and 5.93 (0.77), respectively. All five planned comparisons were significant. P produced higher companionship than N ($T = 2.5$, $p_{adj} < 0.001$, $r_{rb} = 0.987$) and G($T = 23.5$, $p_{adj} < 0.001$, $r_{rb} = 0.814$). Both I ($T = 73.5$, $p_{adj} = 0.015$,$r_{rb} = 0.548$) and D ($T = 9.5$, $p_{adj} < 0.001$, $r_{rb} = 0.937$) produced higher companionship than G. D also produced higher companionship than I ($T = 14.5$, $p_{adj} < 0.001$,$r_{rb} = 0.862$).

Enjoyment. Scores differed significantly across the five configurations ($\chi^2(4) = 24.24$, $p_{adj} < 0.001$, $W = 0.224$). Scores for N, P, G, I, and D were 4.50 (1.32), 5.35 (1.11), 5.02 (1.14), 5.50 (0.80), and 5.69 (0.99). P scored higher than N ($T = 76.5$, $p_{adj} = 0.041$, $r_{rb} = 0.564$), and D scored higher than G ($T = 34$, $p_{adj} = 0.008$, $r_{rb} = 0.731$). The P–G, I–G, and D–I comparisons were not significant (all $p_{adj} \geq 0.074$).

Relaxation. Scores differed significantly across the five configurations ($\chi^2(4) = 16.26$, $p_{adj} = 0.005$, $W = 0.151$). Scores for N, P, G, I, and D were 4.76 (1.34), 5.43 (1.04), 5.09 (0.87), 5.43 (0.74), and 5.39 (0.91). None of the five pairwise comparisons addressing the RQs were significant (all $p_{adj} \geq 0.199$).

Overall Experience. Scores differed significantly across the five configurations ($\chi^2(4) = 28.32$, $p_{adj} < 0.001$, $W = 0.262$). Scores for N, P, G, I, and D were 4.59 (1.28), 5.41 (1.01), 5.30 (0.99), 5.56 (0.85), and 5.85 (0.77). None of the five pairwise comparisons were significant. The adjusted p-values for P–N and D–G were 0.060 and 0.052, all remaining comparisons had $p_{adj} \geq 0.551$.

Objective Knowledge Gain. The overall difference across the five configurations was not significant ($\chi^2(4) = 2.56$, $p_{adj} = 0.634$, $W = 0.024$). Mean gains for N, P, G, I, and D were 1.52 (1.58), 1.07 (1.21), 1.48 (1.12), 1.33 (1.73), and 1.30 (1.81) points. None of the five pairwise comparisons were significant (all $p_{adj} \geq 0.597$).

4.1.3 Role Clarity and Interaction Burden. Role clarity and interaction burden were measured only in configurations I and D. Role clarity scores were 5.12 (0.83) for I and 5.37 (1.13) for D, with no significant difference ($T = 82$, $p_{adj} = 0.302$, $r_{rb} = 0.352$). Interaction burden scores were 3.96 (1.09) for I and 4.10 (1.39) for D, also with no significant difference ($T = 137$, $p_{adj} = 0.502$, $r_{rb} = 0.157$). Cronbach's α for role clarity was 0.751 in I and 0.887 in D, for interaction burden, it was 0.659 and 0.856, respectively.

4.1.4 Configuration Preferences. When asked to select their overall preferred configuration from the five options, 13 participants (48.1%) chose I, followed by D (7, 25.9%), G (3, 11.1%), P (2, 7.4%), and N (1, 3.7%). One participant (3.7%) reported no clear preference.

Five separate pairwise questions compared I and D in terms of overall preference, role clarity, ease of use, interaction burden, and the benefit–burden trade-off. Each question offered three responses: I, D, or "no clear difference".

Only ease of use showed a statistically significant difference after Holm correction: 19 participants (70.4%) found I easier to use, 5 (18.5%) found D easier to use, and 3 (11.1%) reported no clear difference. A two-sided exact binomial test among the 24 participants who selected I or D yielded $p_{adj} = 0.033$.

For overall preference in the I–D comparison, 14 participants selected I, 10 selected D, and 3 reported no clear difference, the difference was not statistically significant ($p_{adj} = 1.000$). Neither role clarity nor the benefit–burden trade-off showed a statistically significant difference (both $p_{adj} = 1.000$). When asked which configuration imposed greater interaction burden, 4 participants selected I, 12 selected D, and 11 reported no clear difference, this difference was also not statistically significant ($p_{adj} = 0.307$).

4.1.5 Interaction Behavior. The logs showed that the mean round durations in N, P, G, I, and D were 2.02, 4.03, 5.21, 4.80, and 6.06 minutes, respectively. N did not provide voice interaction with an Agent. The mean numbers of voice

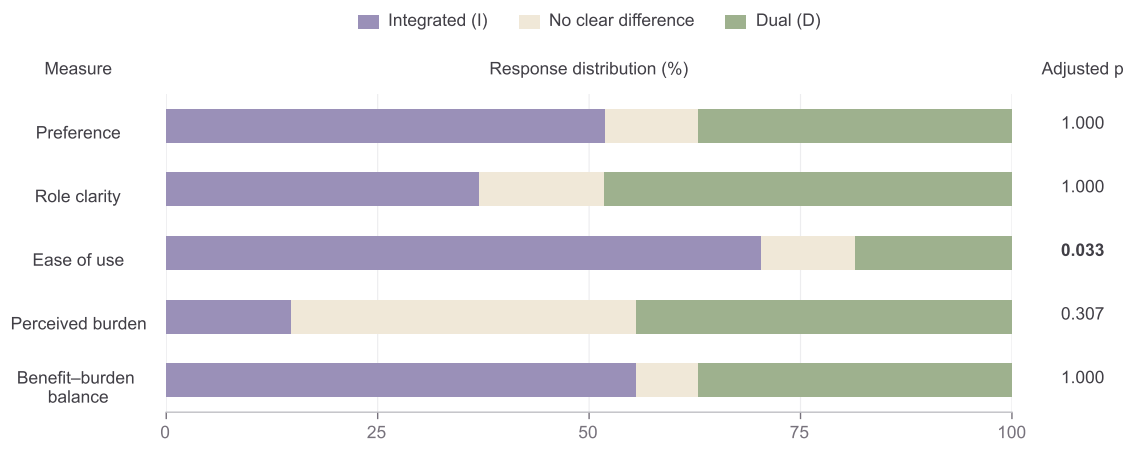


Fig. 5. Participants' comparisons of the Integrated (I) and Dual (D) configurations across five evaluation measures. Stacked bars show the percentage of participants selecting each configuration or reporting no clear difference. For "Perceived burden," selecting a configuration indicates that it was perceived as more burdensome. Adjusted p-values are shown on the right.

submissions in P, G, I, and D were 4.41, 4.56, 4.22, and 5.52, respectively. Voice submissions included supplementary inputs, while round duration included both conversation time and system waiting time.

4.2 Qualitative Results

4.2.1 The Peer Supported Personal Expression and Broadened Perspectives on Artifacts. For some participants, the Peer served as someone with whom they could share personal observations and feelings. When visiting alone, they could only reflect privately on interesting or uncertain aspects of an artifact, whereas *"When you have a Peer visiting with you, you can share your thoughts with her"* (P025). Even when the Peer provided limited specialist knowledge, being able to discuss these thoughts remained valuable. Another participant wanted to explain what they already knew to the Peer and have her act as a listener (P019). In these accounts, participants wanted to contribute their own interpretations as well as receive information. The Peer therefore served as both an audience for and a respondent to their personal expression. The Peer's comments and questions also prompted some participants to examine artifacts from different perspectives. As one participant explained, *"Her comments made me look more carefully, rather than just glancing at the artifact"* (P017). Another participant changed their viewing angle at the Peer's suggestion and then saw the relationship that the Peer had described: *"I changed my viewing angle and found that it really was as she had said"* (P024). This experience made the Peer role feel more realistic. By connecting artifacts with everyday life and personal feelings, the Peer also made them more relevant to participants' own experiences (P011).

However, not all participants wanted or needed alternative perspectives. One participant acknowledged that the Peer offered a different perspective but regarded aesthetic judgment as personal and did not particularly want the Peer to comment on it (P007). Another considered acquiring knowledge itself relaxing and placed greater value on having their curiosity and factual questions addressed (P022). These accounts show that participants differed in the value they placed on personal expression and knowledge acquisition.

4.2.2 Combining Functions Offered Complementary Support but Sometimes Mismatched Participants' Needs. Some participants valued the Integrated Agent because it provided both knowledge and companionship within the same interaction. One participant described the configuration as providing factual information about artifacts while also discussing what it liked about them and why (P018). Another participant viewed companionship as an additional form of support added to the Guide function, *"It essentially added another function without any additional downside"* (P012). For this participant, the Integrated Agent accommodated both knowledge-seeking and the expression of feelings without creating an obvious additional burden. However, other participants felt that combining the two types of content did not match their immediate conversational needs.

One participant recalled that the Integrated Agent seemed to add comments about feelings at the end of a knowledge explanation and described this attempt as *"a little forced"* (P024). Another participant felt that the agent remained focused on explanation, looking for factual points in what they said instead of fully responding to their feelings (P011). One participant also reported receiving factual explanations while seeking emotional companionship (P021). For these participants, receiving both types of content did not necessarily mean receiving an appropriate response.

Some participants also did not clearly perceive the additional companionship support. Because one participant mainly asked artifact-related questions, they perceived little difference between the Integrated Agent and the Guide Agent. In the Dual Agents condition, the same participant also did not notice the Peer taking part in the conversation (P020). Another participant wanted more opportunities to interact with the Peer beyond the Guide's responses (P015). One participant wanted both roles to respond to the same question without having to explicitly mention feelings to prompt a response from the Peer (P024). These accounts complement the variation in Peer participation reported in Section 4.1.4, indicating that some participants experienced only limited companionship support in practice.

4.2.3 Role Separation Clarified Response Expectations but Could Increase Communication Effort. For some participants, separating the roles provided clear recipients for knowledge questions and expressions of feelings, making it easier to anticipate the type of response they would receive. One participant preferred the Dual Agents configuration because its clear division of responsibilities made questions easier to ask. *"I found it easier to ask questions because the two agents had clearly different roles"* (P005). Another participant explained that they could ask the Guide about specialist matters and then share their reactions with the Peer, which simplified how they organized the conversation (P009). Role separation could also reduce the effort required to explain to a single agent whether the user wanted information or companionship (P021). In contrast, other participants felt that the Dual Agents configuration required them to identify the type of request and intended respondent before speaking. One participant felt that they first had to divide their needs into requests for the Guide or the Peer and then specify the corresponding role (P013). Another explained, *"Before saying something, I would think about who would respond"* (P017), making it difficult to combine a question and a feeling naturally within the same response. One participant intended to address the Peer but received a response from the Guide. Before viewing the next artifact, they began adjusting how they formulated their questions: *"I would think about how to phrase it so that the Peer would answer"* (P026). Participants also did not always evaluate role clarity and ease of interaction in the same way. One participant found the separated roles clearer but still considered interacting with a single agent less burdensome (P011). When asked how they distinguished knowledge explanations from responses to feelings in the Integrated Agent condition, another participant said that they had not paid particular attention to this distinction. Instead, they interpreted the agent's replies based on their own questions and expectations, knowledge questions received knowledge-based answers, while expressions of feelings received corresponding responses (P012).

These accounts distinguish understanding an agent's role from obtaining the desired response with little additional communication effort.

4.2.4 The Presence of Two Agents Provided Companionship but Also Introduced Spatial and Social Burdens. For some participants, the simultaneous presence of two agents created a stronger sense of companionship. As one participant explained, *"Visually, there were more people, which made me feel less lonely during the museum visit"* (P026). Another participant expressed a similar view and valued having two conversational partners with distinct functions (P027). A preference for having both agents present did not necessarily indicate satisfaction with their responses. Although the same participant reported that the Peer rarely initiated conversation and replied only a few times, they still preferred having both roles present (P026). At the same time, the presence of two agents could create spatial and psychological pressure. One participant preferred the Integrated Agent because the two agents sometimes blocked their view (P012). Another reported having to consider both agents' locations when moving between artifacts (P018). However, visual obstruction was not unique to the Dual Agents condition. A single agent could also block the view, and one participant found it difficult to adjust to a suitable viewing position (P026). These accounts suggest that spatial inconvenience was related both to the agents' visible presence and to their positioning and the participant's navigation experience. Participants also described social pressure associated with attending to two agents. When repeatedly asking the Guide factual questions, one participant felt that *"the other agent seemed to be ignored"* (P018), which created an additional sense of burden.

5 Discussion

This study compared five agent configurations: No Agent, Peer Agent, Guide Agent, Integrated Agent, and Dual Agents—to examine three issues: the value of Peer companionship in solo VR museum visits, how Peer and Guide functions can be combined, and the benefits and costs of assigning these roles to separate agents. The findings suggest that Peer mainly supported the social side of the museum visit and encouraged visitors to engage with their own interpretations of the exhibits. Adding Peer functions to a Guide could improve the experience, but the outcome depended on whether the combined agent presented a consistent role and adjusted its guidance and companionship to the visitor and the situation. Separating Guide and Peer helped some visitors understand what each agent was responsible for. However, it also required them to choose whom to speak to, anticipate who would respond, and divide their attention between two agents. These findings suggest that museum agents should support flexible roles and forms of participation instead of relying on one fixed configuration.

5.1 The Distinct Value of Peer Companionship in Solo VR Museum Visits

The Peer Agent significantly increased enjoyment compared with No Agent, but offered no advantage over the Guide Agent in perceived knowledge or objective knowledge gain. Its main value was that it gave visitors someone to talk with while allowing them to explore the museum at their own pace. It also gave visitors opportunities to express and develop their own interpretations of the exhibits. We discuss these benefits as companionship without coordination and interpretive participation.

5.1.1 Companionship without Coordination. The interviews further suggest that visiting museums alone does not necessarily mean wanting to be alone[50]. P024 explained that she often visited alone because she wanted to spend more time with individual exhibits. However, she selected Peer Agent as her favorite configuration. P009 similarly explained that visiting alone was often the most practical choice. It allowed the participant to move at a preferred

pace, while friends were not always willing to spend several hours at a museum. These cases show that solo museum visits can result from practical constraints, particularly differences in pace and the difficulty of coordinating time with companions. They do not necessarily reflect a stable preference for solitude. Visiting with another person usually requires coordination of schedules, viewing pace, and route. An AI Peer can be available whenever the visitor wants to interact without determining how long the visitor stays with an exhibit or where the visitor goes next. In this sense, an AI Peer combines the freedom of a solo visit with the option of companionship.

However, solo visitors should not automatically be assumed to want an AI Peer. P007 acknowledged that the Peer could provide a different perspective but considered aesthetic judgment personal. The participant also explained that real companions already understand her interests, aesthetic preferences, and ways of communicating, whereas an AI Peer lacks of this. A simple distinction between visitors who usually visit alone and those who visit with others is therefore unlikely to capture their actual need for AI companionship.

Museum-agent systems should instead consider why visitors choose to visit alone, when they want an agent to participate, and how they want the agent to respond. For example, visitors could choose whether the agent should start conversations, remain quietly present, or speak only when invited. They should also be able to change this preference during the visit. This flexibility would allow an AI Peer to better accommodate differences in how visitors want to experience companionship.

5.1.2 From Social Presence to Interpretive Participation. The value of Peer support extended beyond a general sense of companionship. The interviews show that the Peer also changed how visitors participated in conversations about exhibits. Whereas the Guide mainly provided structured information and positioned visitors as information recipients, the Peer gave visitors someone with whom they could share their feelings, prior knowledge, and interpretations. P019 wanted to tell the Peer what they already knew and have the agent listen. P025 noted that the presence of a Peer made it possible to voice thoughts that would otherwise remain unspoken during a solo visit. These cases show that the value of the Peer lay not only in what it contributed, but also in its ability to listen and respond to visitors' own ideas. The Peer's perspectives could also shape how visitors examined exhibits. P017 explained that when the agent offered a viewpoint, visitors could continue the discussion or express a different opinion. Following a suggestion from the Peer, P024 changed the viewing angle and noticed the relationship between the mother lion and the young lion. In these interactions, the Peer did more than provide factual information. It introduced a perspective that visitors could examine, develop, or challenge. This encouraged closer observation and helped visitors form their own interpretations of the artifacts.

Across these cases, the Peer served at least four functions: a quietly present companion, a listener to visitors' ideas, a discussion starter, and a source of perspectives that encouraged visitors to look again. These functions require different levels of agent initiative. Visitors seeking only a sense of companionship may not want the agent to speak frequently. Those who need a starting point for discussion may prefer the Peer to offer a viewpoint, while those who want to share personal experiences may place greater value on attentive listening and follow-up questions. The design of a Peer role should therefore go beyond using a casual tone or providing affective responses. It should also consider when the agent should participate, how it should respond to visitors' ideas, and how it can leave space for visitors to continue expressing themselves.

These findings further answer RQ1 by showing that an AI Peer can support solo VR museum visits in two ways: by providing companionship without the coordination required for human co-visiting and by supporting visitors' active interpretation of artifacts. The Peer Agent did not provide stronger knowledge support than the Guide Agent or show

an advantage in objective knowledge gain. However, it allowed visitors to express and develop their understanding of exhibits while maintaining their own viewing pace. This interpretation helps explain why Peer Agent increased enjoyment relative to No Agent and clarifies how the Peer role differs from the Guide role.

5.2 Coherent and Adaptive Guide–Peer Integration

5.2.1 Integrating Functions Requires a Coherent Relational Identity. Participants' preferred Guide–Peer ratios reveal different expectations of the Integrated configuration. Among the ten participants who selected Integrated Agent as their overall favorite and provided a clear ratio, six wanted a larger Guide component, three wanted a larger Peer component, and one preferred a 5:5 starting point that could change according to the question. Separately, P011 preferred Peer Agent overall but suggested adding some Guide functions to create "a friend who knows about artifacts". These responses show that choosing Integrated did not mean that participants wanted the same type of agent. Some participants wanted a "friendly guide": an agent that mainly provided accurate and sufficient information but communicated in a more natural way. Others wanted "a friend who knows about artifacts": an agent that primarily interacted as an equal companion but provided expert knowledge when needed. The first design adds a Peer-like communication style to a Guide, whereas the second adds Guide-like knowledge to a Peer. Although both designs include similar functions, they differ in their primary social role and level of authority.

Participants' comments on the current Integrated Agent also show that combining factual and affective content in the same response does not necessarily create a coherent role. P024 felt that the Integrated Agent sometimes added an emotional comment after a factual explanation, making the response feel forced. P011 similarly felt that the agent remained focused on factual explanation and did not fully respond to personal feelings. P021 received a factual explanation while seeking social and emotional support. In these cases, the problem was that the agent's response did not match the visitor's immediate communicative intent. Even when a response contained both factual and affective content, visitors could still perceive the agent as a Guide that occasionally added emotional comments. Perceived knowledgeability and familiarity are therefore useful relational dimensions when designing a coherent combined role [75].

Therefore, the design of an Integrated Agent should prioritize a consistent social role over simply combining functions. "A friend who knows about artifacts" can provide reliable information while maintaining an equal and conversational manner and continuing to engage with the visitor's views. "A friendly guide" can acknowledge the visitor's feelings while remaining primarily responsible for helping the visitor understand the exhibit. A clear and stable role allows visitors to form consistent expectations. Effective integration therefore does not require a fixed division of Guide and Peer content in every response. It requires the same agent to remain consistent and understandable while performing different tasks.

5.2.2 Adapting the Guide–Peer Balance to Museum and Conversational Contexts. An AI agent should maintain a consistent overall role while adjusting the balance between its Guide and Peer functions to the context of the visit. P015 suggested that a Peer may be sufficient in an art gallery, whereas a museum with substantial historical content may require more Guide support. Although our study did not compare museum types, this comment suggests that museum type may shape visitors' needs for background information and open-ended discussion. The underlying factors may include visitors' goals and prior knowledge, how much background information is needed to understand an artifact, and whether visitors are asking for facts, expressing feelings, or inviting discussion. The same visitor may also need different forms of support across artifacts and stages of a visit.

A fixed Guide–Peer balance selected at the beginning of a visit may be too rigid. A more flexible agent could adjust its response to the visitor's current input. When a visitor asks about an artifact's date, material, or historical context, the agent could provide more Guide support. When a visitor expresses a feeling, makes a personal association, or asks for an opinion, the agent could respond more as a Peer. If a response combines a factual question with a personal reaction, the agent could address the visitor's immediate need first and then ask whether additional information would be helpful. Visitors should also be able to adjust the agent's response through simple instructions such as "Tell me more about the background."

Based on these findings, an Integrated Agent could use a two layer design. The first layer would maintain a stable role, such as a "friendly guide" or a "friend who knows about artifacts" to keep the agent's tone, level of initiative, and interaction style consistent. The second layer would adjust the response strategy to the current artifact, question, and visitor reaction. This layer could vary the emphasis on factual explanation, interpretive prompts, affective responses, and proactive questions. The stable role would help visitors understand what kind of agent they were interacting with, while the adaptive response strategy would allow the agent to address changing needs throughout the visit.

These findings answer RQ2 by showing that adding Peer support to Guide-based interaction can expand opportunities for social interaction and artifact interpretation, but its value varies across visitors and contexts. An effective Integrated Agent should maintain a consistent role while adjusting its functional emphasis to the visitor's goals, the characteristics of the artifact, and the visitor's current communicative intent.

5.3 Balancing Role Clarity and Interaction Effort in Separated Guide-Peer Roles

5.3.1 Clearer Roles Do Not Always Make Interaction Easier. The main advantage of the Dual Agents configuration was that it assigned knowledge support and companionship to two agents. P005 found that the clear division of roles made questions easier to ask, while P009 asked the Guide knowledge related questions and shared personal reactions with the Peer. For these participants, role separation provided a clear addressee without requiring them to explain to one agent whether they wanted knowledge or companionship. Having two agents present could also strengthen social presence. P026 and P027 valued having two conversational partners, and for some visitors, the presence of the Peer remained meaningful even when it rarely spoke.

However, these benefits did not lead to a consistent advantage in overall preference or benefit–burden judgments. Understanding the role division and interacting easily are two different issues. Participants could understand what each agent was responsible for but still need to decide whom to address or separate the voice containing both facts and feelings. P011 found the roles clearer but considered a single agent easier to use. The additional effort reported in the Dual Agents condition mainly came from managing two possible respondents. P017 considered who would answer before speaking. P026 expected the Peer to respond but received an answer from the Guide and adjusted the wording of later questions. When the system selected an unexpected respondent, visitors had to make additional decisions and reformulate their input. The Dual Agents condition also had the longest average visit duration and the most voice submissions, but these measures should not be treated as direct evidence of greater engagement.

5.3.2 Making Role Separation Available on Demand. The lack of significant quantitative differences does not mean that Guide and Peer serve similar functions. Rather, these functions do not necessarily need to be embodied by two AI agents that remain present throughout the visit. An Integrated Agent could maintain a continuous conversation while using verbal or interface cues to indicate whether its current response provides knowledge or a Peer perspective. Visitors could also request another perspective or activate a second agent when needed. Role separation would then

become an on-demand feature rather than a continuous interaction burden. For visitors who prefer the Dual Agents configuration, the system should provide more support for conversation management. For example, visitors could select one agent as the primary respondent and invite the other to provide additional responses when needed. They could also adjust each agent's position, visibility, and level of initiative. These controls could preserve role distinction and multi-agent presence while reducing the effort required to select an addressee and divide attention.

Overall, the RQ3 findings show that, for some participants, the Dual Agent configuration clarified agent responsibilities, made the division of functions easier to understand, and enhanced multi-agent presence. However, the evidence is insufficient to show that these benefits generally outweighed the additional interaction costs. A possible design direction is to maintain a continuous single agent interaction while allowing Guide and Peer functions to appear separately when needed. When two agents are present, visitors should be able to manage their participation and spatial positions through simple controls.

5.4 Limitations and Future Work

This study has several limitations. These limitations mainly concern the LLM-driven interaction process, the study duration and experimental design, and the museum context and scope of measurement. Future research should further examine different museum agent configurations in more diverse and longer term settings.

First, all agent responses were dynamically generated by an LLM, and the system selected whether the Guide or Peer should respond in the Dual Agents condition. Human verification provided direct evidence about delivery, routing, and role realization, but also showed that role realization was imperfect: 14 of the 456 delivered replies (3.1%) did not match their assigned roles. In addition, speech-recognition errors, response latency, and variation in response quality may also have influenced participants' evaluations. Replications with different language models, voices, and virtual embodiments are needed to establish the robustness of the findings.

Second, we used a within-participant design in which each participant experienced five agent configurations in a single experimental session. Each museum visit lasted approximately five minutes, and participants were required to ask at least three questions. Although Williams sequences were used to counterbalance condition order and exhibit themes, carryover effects may still have occurred. Requiring participants to initiate questions also differed from an autonomous museum visit. Future research could conduct longer and more natural visits using between participant or longitudinal designs. Allowing visitors to decide whether, when, and how to interact with the agents would help determine whether their needs and preferences change over time.

Finally, this study was conducted in a VR museum focused on porcelain artifacts, and the findings may not generalize directly to other types of museums. Our evaluation primarily captured participants' immediate subjective experiences and short-term interaction behaviors, without examining continued use intentions or long-term companionship experiences. Future studies should investigate a broader range of museum contexts and introduce richer behavioral measures.

6 Conclusion

This study examined whether an AI Peer can support companionship during solo VR museum visits and how the Peer and Guide roles should be configured. We designed and compared five configurations: No Agent, Peer Agent, Guide Agent, Integrated Agent, and Dual Agents. We conducted a within-participant study with 27 participants to evaluate their effects on visitor experience, knowledge acquisition, companionship, and interaction burden.

The results show that the Peer increased enjoyment compared with No Agent. Its main value was providing companionship and encouraging visitors to express and develop their own interpretations rather than improving knowledge acquisition. Adding Peer functions to a Guide did not consistently improve all measures. The value of the Integrated configuration depended on whether the agent maintained a coherent role and adapted its Guide and Peer functions to visitors' needs. Separating the Guide and Peer roles helped some participants understand each agent's responsibilities, but also increased the effort required to select a respondent, divide attention, and manage multiple agents. These findings indicate that no single configuration is optimal for all visitors and museum contexts.

Companionship should therefore be treated as a design goal distinct from knowledge guidance. Museum agents should maintain clear and coherent roles while flexibly adapting their Guide and Peer functions to visitors' immediate needs. Role separation could also be provided on demand rather than used as a fixed configuration.

References

[1] Amy L Baylor and Yanghee Kim. 2005. Simulating instructional roles through pedagogical agents. *International Journal of Artificial intelligence in education* 15, 2 (2005), 95–115.

[2] Tessa Beinema, Daniel Davison, Dennis Reidsma, Oresti Banos, Merijn Bruijnes, Brice Donval, Álvaro Fides Valero, Dirk Heylen, Dennis Hofs, Gerwin Huizing, et al. 2021. Agents United: An open platform for multi-agent conversational systems. In *Proceedings of the 21st ACM international conference on intelligent virtual agents*. 17–24.

[3] Timothy Bickmore and Justine Cassell. 2001. Relational agents: a model and implementation of building user trust. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 396–403.

[4] Timothy Bickmore, Laura Pfeifer, and Daniel Schulman. 2011. Relational agents improve engagement and learning in science museum visitors. In *International Workshop on Intelligent Virtual Agents*. Springer, 55–67.

[5] Timothy W Bickmore and Rosalind W Picard. 2005. Establishing and maintaining long-term human-computer relationships. *ACM Transactions on Computer-Human Interaction (TOCHI)* 12, 2 (2005), 293–327.

[6] Timothy W Bickmore, Laura M Pfeifer Vardoulakis, and Daniel Schulman. 2013. Tinker: a relational agent museum guide. *Autonomous agents and multi-agent systems* 27, 2 (2013), 254–276.

[7] Frank Biocca, Chad Harms, and Jenn Gregg. 2001. The Networked Minds Measure of Social Presence: Pilot Test of the Factor Structure and Concurrent Validity. In *Proceedings of the Fourth Annual International Workshop on Presence*. International Society for Presence Research, Philadelphia, PA, USA, 1–9.

[8] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.

[9] Barry Brown, Ian MacColl, Matthew Chalmers, Areti Galani, Cliff Randell, and Anthony Steed. 2003. Lessons from the lighthouse: collaboration in a shared mixed reality system. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 577–584.

[10] Alberto Cannavò, Simona Pacchiotti, Nicola Retta, Martina Terzoli, Roberta Spallone, and Fabrizio Lamberti. 2024. Passive haptics and conversational avatars for interacting with ancient Egypt remains in high-fidelity virtual reality experiences. *ACM Journal on Computing and Cultural Heritage* 17, 2 (2024), 1–28.

[11] Justine Cassell. 2001. Embodied conversational agents: representation and intelligence in user interfaces. *AI magazine* 22, 4 (2001), 67–83.

[12] Rijul Chaturvedi, Sanjeev Verma, Ronnie Das, and Yogesh K Dwivedi. 2023. Social companionship with artificial intelligence: Recent trends and future avenues. *Technological Forecasting and Social Change* 193 (2023), 122634.

[13] Bingqing Chen, Wenqi Chu, Xubo Yang, and Yue Li. 2025. Voice of artifacts: Evaluating user preferences for artifact voice in VR museums. *Computers & Graphics* 133 (2025), 104473.

[14] Dimitra Christidou. 2018. Art on the move: The role of joint attention in visitors' encounters with artworks. *Learning, Culture and Social Interaction* 19 (2018), 1–10.

[15] Seung Jin Chung, So Yeon Kim, and Ki Han Kim. 2024. Comparison of visitor experiences of virtual reality exhibitions by spatial environment. *International Journal of Human-Computer Studies* 181 (2024), 103145.

[16] Eugene Ch'Ng, Yue Li, Shengdan Cai, and Fui-Theng Leow. 2020. The effects of VR environments on the acceptance, experience, and expectations of cultural heritage learning. *Journal on computing and cultural heritage (JOCCH)* 13, 1 (2020), 1–21.

[17] Gabriele Cierniak, Katharina Scheiter, and Peter Gerjets. 2009. Explaining the split-attention effect: Is the reduction of extraneous cognitive load accompanied by an increase in germane cognitive load? *Computers in Human Behavior* 25, 2 (2009), 315–324.

[18] Lee J Cronbach. 1951. Coefficient alpha and the internal structure of tests. *psychometrika* 16, 3 (1951), 297–334.

[19] Andrea Dordio, Eva Lancho, María José Merchán, and Pilar Merchán. 2024. Cultural heritage as a didactic resource through extended reality: A systematic review of the literature. *Multimodal Technologies and Interaction* 8, 7 (2024), 58.

[20] John H Falk. 2009. *Identity and the museum visitor experience*. Left Coast Press.

[21] John H Falk and Lynn D Dierking. 2013. *Museum experience revisited.* Left Coast Press.

[22] Milton Friedman. 1937. The use of ranks to avoid the assumption of normality implicit in the analysis of variance. *Journal of the american statistical association* 32, 200 (1937), 675–701.

[23] Irene Lopez Garcia, Ephraim Schott, Marcel Gohsen, Volker Bernhard, Benno Stein, and Bernd Froehlich. 2024. Speaking with objects: Conversational agents' embodiment in virtual museums. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR).* IEEE, 279–288.

[24] Anne-Britt Gran, Nina Lager Vestberg, Peter Booth, and Anne Ogundipe. 2019. A digital museum's contribution to diversity–a user study. *Museum management and curatorship* 34, 1 (2019), 58–78.

[25] Manuel Guimarães, Rui Prada, Pedro A Santos, João Dias, Arnav Jhala, and Samuel Mascarenhas. 2020. The impact of virtual reality in the social presence of a virtual agent. In *Proceedings of the 20th ACM international conference on intelligent virtual agents.* 1–8.

[26] Taicheng Guo, Xiuying Chen, Yaqi Wang, Ruidi Chang, Shichao Pei, Nitesh V Chawla, Olaf Wiest, and Xiangliang Zhang. 2024. Large language model based multi-agents: A survey of progress and challenges. *arXiv preprint arXiv:2402.01680* (2024).

[27] George E Hein. 2002. *Learning in the Museum.* routledge.

[28] Sture Holm. 1979. A simple sequentially rejective multiple test procedure. *Scandinavian journal of statistics* 6, 2 (1979), 65–70.

[29] Eva Hornecker and Luigina Ciolfi. 2022. *Human-computer interactions in museums.* Springer Nature.

[30] Wolfgang Hürst, Ferdinand de Coninck, and Xhi Jia Tan. 2016. Complementing artworks to create immersive VR museum experiences. In *Proceedings of the 13th international conference on advances in computer entertainment technology.* 1–6.

[31] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of hallucination in natural language generation. *ACM computing surveys* 55, 12 (2023), 1–38.

[32] W Lewis Johnson and James C Lester. 2018. Pedagogical agents: Back to the future. *AI Magazine* 39, 2 (2018), 33–44.

[33] Reshmashree B Kantharaju, Dominic De Franco, Alison Pease, and Catherine Pelachaud. 2018. Is two better than one? Effects of multiple agents on user persuasion. In *Proceedings of the 18th international conference on intelligent virtual agents.* 255–262.

[34] M. G. Kendall and B. Babington Smith. 1939. The Problem of m Rankings. *The Annals of Mathematical Statistics* 10, 3 (1939), 275–287. doi:10.1214/aoms/1177732186

[35] Dave S Kerby. 2014. The simple difference formula: An approach to teaching nonparametric correlation. *Comprehensive Psychology* 3 (2014), 11–IT.

[36] Yanghee Kim, Amy L Baylor, and Pals Group. 2006. Pedagogical agents as learning companions: The role of agent competency and type of interaction. *Educational technology research and development* 54, 3 (2006), 223–243.

[37] Kalevi M Korpela, Matti Ylén, Liisa Tyrväinen, and Harri Silvennoinen. 2008. Determinants of restorative experiences in everyday favorite places. *Health & place* 14, 4 (2008), 636–652.

[38] Gaea Leinhardt, Kevin Crowley, and Karen Knutson. 2003. *Learning conversations in museums.* Taylor & Francis.

[39] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems* 33 (2020), 9459–9474.

[40] Yue Li, Eugene Ch'ng, and Sue Cobb. 2023. Factors influencing engagement in hybrid virtual and augmented reality. *ACM Transactions on Computer-Human Interaction* 30, 4 (2023), 1–27.

[41] Yue Li, Paul Tennent, and Sue Cobb. 2018. Appropriate control methods for mobile virtual exhibitions. In *International Conference on VR Technologies in Cultural Heritage.* Springer, 165–183.

[42] Tze Wei Liew, Su-Mae Tan, and Chandrika Jayothisa. 2013. The effects of peer-like and expert-like pedagogical agents on learners' agent perceptions, task-related attitudes, and learning achievement. *Journal of Educational Technology & Society* 16, 4 (2013), 275–286.

[43] Kate Loveys, Catherine Hiko, Mark Sagar, Xueyuan Zhang, and Elizabeth Broadbent. 2022. "I felt her company": A qualitative study on factors affecting closeness and emotional support seeking with an embodied conversational agent. *International Journal of Human-Computer Studies* 160 (2022), 102771.

[44] Richard E Mayer, Kristina Sobko, and Patricia D Mautone. 2003. Social cues in multimedia learning: Role of speaker's voice. *Journal of educational Psychology* 95, 2 (2003), 419.

[45] Paulette M McManus. 1987. It's the company you keep…: The social determination of learning-related behaviour in a science museum. *Museum Management and Curatorship* 6, 3 (1987), 263–270.

[46] Clifford Nass and Youngme Moon. 2000. Machines and mindlessness: Social responses to computers. *Journal of social issues* 56, 1 (2000), 81–103.

[47] Nahal Norouzi, Kangsoo Kim, Jason Hochreiter, Myungho Lee, Salam Daher, Gerd Bruder, and Greg Welch. 2018. A systematic survey of 15 years of user studies published in the intelligent virtual agents conference. In *Proceedings of the 18th international conference on intelligent virtual agents.* 17–22.

[48] David Novick, Mahdokht Afravi, Adriana Camacho, Aaron Rodriguez, and Laura Hinojos. 2019. Pedagogical-agent learning companions in a virtual reality educational experience. In *International Conference on Human-Computer Interaction.* Springer, 193–203.

[49] Mohd Kamal Othman, Helen Petrie, and Christopher Power. 2011. Engaging visitors in museums with technology: scales for the measurement of visitor and multimedia guide experience. In *IFIP conference on human-computer interaction.* Springer, 92–99.

[50] Jan Packer and Roy Ballantyne. 2005. Solitary vs. shared: Exploring the social dimension of museum learning. *Curator: The Museum Journal* 48, 2 (2005), 177–192.

[51] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. Robust speech recognition via large-scale weak supervision. In *International conference on machine learning*. PMLR, 28492–28518.

[52] Byron Reeves and Clifford Nass. 1996. The media equation: How people treat computers, television, and new media like real people. *Cambridge, UK* 10, 10 (1996), 19–36.

[53] Minjin Rheu, Ji Youn Shin, Wei Peng, and Jina Huh-Yoo. 2021. Systematic review: Trust-building factors and implications for conversational agent design. *International Journal of Human–Computer Interaction* 37, 1 (2021), 81–96.

[54] Jeff Rickel and WL Johnson. 2000. Task-oriented collaboration with embodied agents. *Embodied conversational agents* (2000), 95.

[55] Rufat Rzayev, Florian Habler, Polina Ugnivenko, Niels Henze, and Valentin Schwind. 2020. It's Not Always Better When We're Together: Effects of Being Accompanied in Virtual Reality. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–8.

[56] Rufat Rzayev, Gürkan Karaman, Niels Henze, and Valentin Schwind. 2019. Fostering virtual guide in exhibitions. In *Proceedings of the 21st international conference on human-computer interaction with mobile devices and services*. 1–6.

[57] Briane Paul V Samson and Yasuyuki Sumi. 2020. Are two heads better than one? Exploring two-party conversations for car navigation voice guidance. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–9.

[58] Ricarda Schlimbach, Heidi Rinn, Daniel Markgraf, and Susanne Robra-Bissantz. 2022. A literature review on pedagogical conversational agent adaptation. (2022).

[59] Luyao Shen, Xian Wang, Sijia Li, Lik-Hang Lee, Mingming Fan, and Pan Hui. 2025. EmojiChat: Toward designing emoji-driven social interaction in VR museums. *International Journal of Human–Computer Interaction* 41, 11 (2025), 6987–7003.

[60] Paweł M Strojny, Natalia Dużmańska-Misiarczyk, Natalia Lipp, and Agnieszka Strojny. 2020. Moderators of social facilitation effect in virtual reality: Co-presence and realism of virtual agents. *Frontiers in psychology* 11 (2020), 1252.

[61] Mingyang Su, Chao Liu, Jingling Zhang, WU Shuang, and Mingming Fan. 2025. Simviews: An interactive multi-agent system simulating visitor-to-visitor conversational patterns to present diverse perspectives of artifacts in virtual museums. In *Proceedings of the 33rd ACM International Conference on Multimedia*. 6740–6750.

[62] John Sweller. 1988. Cognitive load during problem solving: Effects on learning. *Cognitive science* 12, 2 (1988), 257–285.

[63] Stella Sylaiou, Katerina Mania, Athanasis Karoulis, and Martin White. 2010. Exploring the relationship between presence and enjoyment in a virtual museum. *International journal of human-computer studies* 68, 5 (2010), 243–253.

[64] David Traum and Jeff Rickel. 2002. Embodied agents for multi-party dialogue in immersive virtual worlds. In *Proceedings of the first international joint conference on Autonomous agents and multiagent systems: part 2*. 766–773.

[65] Matias Volonte, Eyal Ofek, Ken Jakubzak, Shawn Bruner, and Mar Gonzalez-Franco. 2022. Headbox: A facial blendshape animation toolkit for the microsoft rocketbox library. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. IEEE, 39–42.

[66] Dirk Vom Lehn, Christian Heath, and Jon Hindmarsh. 2001. Exhibiting interaction: Conduct and collaboration in museums and galleries. *Symbolic interaction* 24, 2 (2001), 189–216.

[67] Xueqi Wang, Yue Li, and Hai-Ning Liang. 2024. Magicmap: Enhancing indoor navigation experience in vr museums. In *2024 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*. IEEE, 881–891.

[68] Xueqi Wang, Yue Li, Boge Ling, Han-Mei Chen, and Hai-Ning Liang. 2025. ResponsiveView: Enhancing 3D Artifact Viewing Experience in VR Museums. *IEEE Transactions on Visualization and Computer Graphics* 31, 5 (2025), 2870–2879.

[69] Zhiyang Wang, Jina Lee, and Stacy Marsella. 2013. Multi-party, multi-role comprehensive listening behavior. *Autonomous Agents and Multi-Agent Systems* 27, 2 (2013), 218–234.

[70] Zhan Wang, Lin-Ping Yuan, Liangwei Wang, Bingchuan Jiang, and Wei Zeng. 2024. Virtuwander: Enhancing multi-modal interaction for virtual tour guidance through large language models. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–20.

[71] Frank Wilcoxon. 1945. Individual comparisons by ranking methods. *Biometrics bulletin* 1, 6 (1945), 80–83.

[72] Evan James Williams. 1949. Experimental designs balanced for the estimation of residual effects of treatments. *Australian Journal of Scientific Research Series A: Physical Sciences* 2, 2 (1949), 149–168.

[73] Fu-Chia Yang, Minsoo Choi, Dominic Kao, and Christos Mousas. 2026. On the Intelligence and Knowledgeability of Virtual Agents. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*. 1–21.

[74] Fu-Chia Yang, Kevin Duque, and Christos Mousas. 2024. The effects of depth of knowledge of a virtual agent. *IEEE Transactions on Visualization and Computer Graphics* 30, 11 (2024), 7140–7151.

[75] Fu-Chia Yang, Siqi Guo, and Christos Mousas. 2025. Exploring familiarity and knowledgeability in conversational virtual agents. *ACM Transactions on Applied Perception* 23, 1 (2025), 1–28.

[76] Zihao Zhu, Ao Yu, Xin Tong, and Pan Hui. 2025. Exploring LLM-Powered Role and Action-Switching Pedagogical Agents for History Education in Virtual Reality. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–19.